

# REUBEN SAMSON RAJ

rs077@uark.edu · +1-479-332-8001 · Fayetteville, AR

[Website](#) · [LinkedIn](#) · [Google Scholar](#)

## EDUCATION

|                                                                                                                                                                                                                                       |                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Ph.D. in Computer Science</b> , University of Arkansas, Fayetteville, AR<br>Dissertation: <i>Frameworks on Network Programmability and Abstraction for Securing and Strengthening Smart-Grids</i><br>Advisor: Dr. Dong (Kevin) Jin | Aug 2021 – Dec 2026 (Expected) |
| <b>M.S. in Computer Engineering</b> , University of Arkansas, Fayetteville, AR                                                                                                                                                        | Aug 2021 – Dec 2024            |
| <b>B.E. in ECE</b> , College of Engineering, Guindy, Anna University, Chennai, India                                                                                                                                                  | Jun 2006 – May 2010            |

## RESEARCH INTERESTS

Network Security, Programmable Networks, Smart-Grid/ICS Security, Cloud-Native Systems

## WORK EXPERIENCE

|                                                                         |                                         |
|-------------------------------------------------------------------------|-----------------------------------------|
| <b>Sr. Graduate Research Assistant</b><br><i>University of Arkansas</i> | Aug 2021 – Present<br>Fayetteville, AR  |
| <b>Network Architecture Research Intern</b><br><i>Nokia Bell Labs</i>   | Jun 2023 – Aug 2023<br>Murray Hill, NJ  |
| <b>Software Engineer</b><br><i>Cisco Systems</i>                        | Jun 2020 – Jul 2021<br>Bangalore, India |
| <b>Member of Technical Staff</b><br><i>Parallel Wireless</i>            | Oct 2018 – May 2020<br>Bangalore, India |
| <b>R&amp;D Engineer</b><br><i>Nokia Networks</i>                        | Apr 2015 – Oct 2018<br>Bangalore, India |
| <b>Software Engineer</b><br><i>Cisco Systems</i>                        | Apr 2012 – Apr 2015<br>Bangalore, India |
| <b>Associate Software Engineer</b><br><i>Accenture</i>                  | Apr 2011 – Mar 2012<br>Chennai, India   |

## PUBLICATIONS

- Reuben Samson Raj**, Dong Jin. *A Framework to Evaluate PMU Networks for Resiliency Under Network Failure Conditions*. IEEE SmartGridComm 2022 [🔗](#)
- Zhiyao He, Yanfeng Qu, Gong Chen, **Reuben Samson Raj**, Hui Lin, Dong Jin. *Towards Secure and Resilient Synchrophasor Networks Using P4 Programmable Switches*. IEEE GreenTech 2024 [🔗](#)
- Reuben Samson Raj**, Dong Jin. *Dynamic Data Driven Security Framework for Industrial Control Networks using Programmable Switches*. DDDAS 2024 [🔗](#)
- Reuben Samson Raj**, Dong Jin. *Leveraging Data Plane Programmability Towards a Policy-driven In-Network Security Framework for Industrial Control Systems*. ACM e-Energy 2025 [🔗](#)
- Reuben Samson Raj**, Dong Jin. *A Dynamic Data-Driven Framework for Enhancing Energy System Security and Resilience through Programmable Networks*. Book Chapter - DDDAS Handbook Vol. IV, Springer (Accepted-Under Production) [🔗](#)

6. Sarvesh Bidkar, Rakesh Abbireddy, Maryam Amiri, **Reuben Samson Raj**, Anil Rana, Jeff McLaird, Paul Rea, Meryem Simsek, Jesse E. Simsarian. *Demonstration of Multi-Provider Network and Cloud Service Provisioning with Blockchain Smart Contracts*. ECOC 2025 [🔗](#)
7. Zhiyao He\*, **Reuben Samson Raj**\*, Yanfeng Qu, Dong Jin. *Programmable Data Plane Approaches for ML-Based Anomalous Event Classification in Grid Sensor Networks*. IEEE ICC 2026 (\* - Equal Contributions) [🔗](#)
8. Yuexin Chen, **Reuben Samson Raj**, Xiaoyu Guo, Dong Jin. *Enhancing Security of Grid Control Systems through In-network Collaboration of Programmable Data Planes*. IEEE SmartGridComm 2026 (Accepted) [🔗](#)
9. Haden Fowler, **Reuben Samson Raj**, Dong Jin. *Dynamic Data-Driven Cyber Defense for Industrial Control Systems using P4-Based Cross-Layer Telemetry*. DDDAS 2026 (Accepted)

## TEACHING EXPERIENCE

---

|                                                                                 |           |
|---------------------------------------------------------------------------------|-----------|
| <b>CSCE 47503 Computer Networks</b> — Substitute Instructor                     | Fall 2025 |
| <b>CSCE 47503 Computer Networks</b> — Teaching Assistant, Substitute Instructor | Fall 2026 |

## AWARDS & HONORS

---

|                                                                         |      |
|-------------------------------------------------------------------------|------|
| • 2nd place - Best Paper Award, <b>DDDAS Conference</b>                 | 2024 |
| • Student Travel Grant, <b>ACM e-Energy</b>                             | 2025 |
| • Larry & Gwen Stephens Graduate Scholarship                            | 2024 |
| • Reginald R. Barney & Jameson A. Baxter Graduate Scholarship           | 2023 |
| • Department of Electrical Engineering and Computer Science Scholarship | 2022 |
| • You Amaze and You Inspire Awards, <b>Cisco</b>                        | 2020 |
| • Best Feature Team Award, <b>Nokia</b>                                 | 2016 |
| • CAP Award, <b>Cisco</b>                                               | 2013 |

## ACADEMIC SERVICE & VOLUNTEERING

---

- Reviewer, **ACM TOMACS**, 2025, 2026
- Reviewer, **IEEE SmartGridComm**, 2026
- Mentoring and guidance for undergraduate honor's thesis projects
- Served as vendor liaison and key coding platform point-of-contact for regional [high-school programming contests](#)

## SKILLS

---

|                                  |                                                          |
|----------------------------------|----------------------------------------------------------|
| <b>Languages</b>                 | Python, P4 (Tofino, bmv2), C, Bash                       |
| <b>Networking</b>                | SDN, P4, 5G Core, EPC, TCP/IP, Mininet, Wireshark, iperf |
| <b>Cloud &amp; Orchestration</b> | Kubernetes, Docker, OpenStack                            |
| <b>Security &amp; ICS/OT</b>     | Modbus, DNP3, Network Security, IDA Pro, PE Studio       |
| <b>Tools &amp; Frameworks</b>    | Linux, Git, FastAPI, OpenCV, NVIDIA GPU Stack            |

## MEDIA REPORTS

---

[Second Best Paper in DDDAS 2024](#)